

```
mysql> show databases;
```

| Database           |
|--------------------|
| cg_mysql           |
| information_schema |
| mysql              |
| performance_schema |
| sys                |

```
5 rows in set (0.04 sec)
```

```
mysql> use cg_mysql;
```

```
Database changed
```

```
mysql> /* MySQL DCL Commands */
```

```
mysql> /* DCL commands are used to control access to data in the database - mainly  
to grant and revoke privileges to users. */
```

```
mysql> /* GRANT = gives specific privileges to a user or role
```

```
/*>    REVOKE = takes back the privileges from a user or role
```

```
/*>
```

```
/*> Common privileges you can grant:
```

```
/*> -> select = read data
```

```
/*> -> insert = insert data
```

```
/*> -> update = modify data
```

```
/*> -> delete = delete data
```

```
/*> -> all privileges = all the privileges
```

```
/*>
```

```
/*> User Creation: create user "username" identified by "password"
```

```
/*>
```

```
/*> grant privilege: grant privilege on table_name to "username"
```

```
/*> revoke privilege: revoke privilege on table_name to "username" */
```

```
mysql> show tables;
```

| Tables_in_cg_mysql |
|--------------------|
| company_staff      |
| departments        |
| doctors            |
| employees          |
| staff              |

```
5 rows in set (0.07 sec)
```

```
mysql> create table students(
```

```
-> student_id int primary key,
```

```
-> student_name varchar(30),
```

```
-> student_email varchar(20) unique
```

```
-> );
```

```
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> insert into values(101,'Praveen','praveen@cg.com');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MySQL server version for the right syntax to use near
'values(101,'Praveen','praveen@cg.com')' at line 1
mysql> insert into students values(101,'Praveen','praveen@cg.com');
Query OK, 1 row affected (0.07 sec)
```

```
mysql> select * from students;
+-----+-----+-----+
| student_id | student_name | student_email |
+-----+-----+-----+
|          101 | Praveen      | praveen@cg.com |
+-----+-----+-----+
1 row in set (0.04 sec)
```

```
mysql> insert into students
values(102,'jayakumar','jayakumar@cg.com'),(103,'sanjay','sanjay@cg.com');
Query OK, 2 rows affected (0.06 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> select * from students;
+-----+-----+-----+
| student_id | student_name | student_email |
+-----+-----+-----+
|          101 | Praveen      | praveen@cg.com |
|          102 | jayakumar    | jayakumar@cg.com |
|          103 | sanjay       | sanjay@cg.com |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> create user jk identified by 'jk123';
ERROR 1396 (HY000): Operation CREATE USER failed for 'jk'@'%'
mysql> create user vjak identified by 'jk123';
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> create user vuss identified by 'us123';
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> grant select, insert, update on students to vjak;
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> grant select, drop on students to vuss;
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> select * from students;
+-----+-----+-----+
| student_id | student_name | student_email |
+-----+-----+-----+
|          101 | Praveen      | praveen@cg.com |
|          102 | jayakumar    | jayakumar@cg.com |
+-----+-----+-----+
```

|     |        |               |
|-----|--------|---------------|
| 103 | sanjay | sanjay@cg.com |
| 104 | usha   | usha@cg.com   |

4 rows in set (0.00 sec)

```
mysql> select * from students;
```

| student_id | student_name | student_email    |
|------------|--------------|------------------|
| 101        | Praveen      | praveen@cg.com   |
| 102        | jayakumar    | jayakumar@cg.com |
| 103        | sanjay       | sanjay@cg.com    |
| 104        | ushasri      | usha@cg.com      |

4 rows in set (0.00 sec)

```
mysql> revoke insert, update on students from vjak;
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> revoke drop on students from vuss;
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> create table emp(
  -> emp_id int auto_increment primary key,
  -> ename varchar(30),
  -> esalary decimal(10,2)
  -> );
Query OK, 0 rows affected (0.09 sec)
```

```
mysql> insert into emp values('jayakumar',50000.00),('praveen',40000.00);
ERROR 1136 (21S01): Column count doesn't match value count at row 1
mysql> insert into emp(ename,esalary)
values('jayakumar',50000.00),('praveen',40000.00);
Query OK, 2 rows affected (0.06 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql> select * from emp;
```

| emp_id | ename     | esalary  |
|--------|-----------|----------|
| 1      | jayakumar | 50000.00 |
| 2      | praveen   | 40000.00 |

2 rows in set (0.00 sec)

```
mysql> delete emp where emp_id = 2;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MySQL server version for the right syntax to use near 'where
emp_id = 2' at line 1
mysql> delete table emp where emp_id = 2;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'table emp where emp\_id = 2' at line 1

mysql> delete from emp where emp\_id = 2;

Query OK, 1 row affected (0.04 sec)

mysql> select \* from emp;

| emp_id | ename     | esalary  |
|--------|-----------|----------|
| 1      | jayakumar | 50000.00 |

1 row in set (0.00 sec)

mysql> insert into emp(ename,esalary) values('sajay',30000.00);

Query OK, 1 row affected (0.04 sec)

mysql> select \* from emp;

| emp_id | ename     | esalary  |
|--------|-----------|----------|
| 1      | jayakumar | 50000.00 |
| 3      | sajay     | 30000.00 |

2 rows in set (0.00 sec)

mysql> create table backup\_emp as

-> select \* from emp;

Query OK, 2 rows affected (0.10 sec)

Records: 2 Duplicates: 0 Warnings: 0

mysql> select \* from backup\_emp;

| emp_id | ename     | esalary  |
|--------|-----------|----------|
| 1      | jayakumar | 50000.00 |
| 3      | sajay     | 30000.00 |

2 rows in set (0.00 sec)

mysql> -- Computed columns

mysql> alter table emp add exp int;

Query OK, 0 rows affected (0.08 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> alter table emp add t\_earning decimal generated always as (esalary \* exp);

Query OK, 0 rows affected (0.08 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> select \* from emp;

| emp_id | ename     | esalary  | exp  | t_earning |
|--------|-----------|----------|------|-----------|
| 1      | jayakumar | 50000.00 | NULL | NULL      |
| 3      | sajay     | 30000.00 | NULL | NULL      |

2 rows in set (0.00 sec)

```
mysql> insert into emp(exp) values(24),(60) where empid = 1,3;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MySQL server version for the right syntax to use near 'where
empid = 1,3' at line 1
```

```
mysql> insert into emp(exp) values(24) where emp_id = 1;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MySQL server version for the right syntax to use near 'where
emp_id = 1' at line 1
```

```
mysql> insert into emp(exp) values(24),(60);
```

Query OK, 2 rows affected (0.04 sec)

Records: 2 Duplicates: 0 Warnings: 0

```
mysql> select * from emp;
```

| emp_id | ename     | esalary  | exp  | t_earning |
|--------|-----------|----------|------|-----------|
| 1      | jayakumar | 50000.00 | NULL | NULL      |
| 3      | sajay     | 30000.00 | NULL | NULL      |
| 4      | NULL      | NULL     | 24   | NULL      |
| 5      | NULL      | NULL     | 60   | NULL      |

4 rows in set (0.00 sec)

```
mysql> delete from emp where emp_id = 4;
```

Query OK, 1 row affected (0.01 sec)

```
mysql> delete from emp where emp_id = 5;
```

Query OK, 1 row affected (0.05 sec)

```
mysql> update emp set exp = 24 where emp_id = 1;
```

Query OK, 1 row affected (0.06 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
mysql> update emp set exp = 60 where emp_id = 3;
```

Query OK, 1 row affected (0.07 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
mysql> select * from emp;
```

| emp_id | ename     | esalary  | exp | t_earning |
|--------|-----------|----------|-----|-----------|
| 1      | jayakumar | 50000.00 | 24  | 1200000   |

|   |       |          |    |         |
|---|-------|----------|----|---------|
| 3 | sajay | 30000.00 | 60 | 1800000 |
|---|-------|----------|----|---------|

2 rows in set (0.00 sec)

```
mysql> alter table emp drop column t_earning;
Query OK, 0 rows affected (0.12 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> select * from emp;
```

| emp_id | ename     | esalary  | exp |
|--------|-----------|----------|-----|
| 1      | jayakumar | 50000.00 | 24  |
| 3      | sajay     | 30000.00 | 60  |

2 rows in set (0.00 sec)

```
mysql> alter table emp add t_earning decimal(10,2) generated always as (esalary *
exp);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> select * from emp;
```

| emp_id | ename     | esalary  | exp | t_earning  |
|--------|-----------|----------|-----|------------|
| 1      | jayakumar | 50000.00 | 24  | 1200000.00 |
| 3      | sajay     | 30000.00 | 60  | 1800000.00 |

2 rows in set (0.00 sec)

```
mysql> notee
```

```
=====
user = vjak
=====
```

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u vjak -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.36 MySQL Community Server - GPL
```

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> show databases;
+-----+
| Database |
+-----+
| cg_mysql |
| information_schema |
| performance_schema |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> use cg_mysql;
Database changed
mysql> show tables;
+-----+
| Tables_in_cg_mysql |
+-----+
| students |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select * from students;
+-----+-----+-----+
| student_id | student_name | student_email |
+-----+-----+-----+
| 101 | Praveen | praveen@cg.com |
| 102 | jayakumar | jayakumar@cg.com |
| 103 | sanjay | sanjay@cg.com |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> insert into students values(104,'usha','usha@cg.com');
Query OK, 1 row affected (0.07 sec)
```

```
mysql> delete from students where student_name = sanjay;
ERROR 1142 (42000): DELETE command denied to user 'vjak'@'localhost' for table 'students'
```

```
mysql> update table students set student_name = 'ushasri' where student_name = 'usha';
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'table students set student_name = 'ushasri' where student_name = 'usha'' at line 1
```

```
mysql> update students set student_name = 'ushasri' where student_name = 'usha';
Query OK, 1 row affected (0.04 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from students;
+-----+-----+-----+
| student_id | student_name | student_email |
+-----+-----+-----+
```

|     |           |                  |
|-----|-----------|------------------|
| 101 | Praveen   | praveen@cg.com   |
| 102 | jayakumar | jayakumar@cg.com |
| 103 | sanjay    | sanjay@cg.com    |
| 104 | ushasri   | usha@cg.com      |

4 rows in set (0.00 sec)

```
mysql> insert into students values(105,'pavankalyan','pavan@cg.com')
```

```
-> ;
```

```
ERROR 1142 (42000): INSERT command denied to user 'vjak'@'localhost' for table 'students'
```

```
mysql>
```

```
=====
```

```
user = vuss
```

```
=====
```

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u vuss -p
```

```
Enter password: *****
```

```
Welcome to the MySQL monitor. Commands end with ; or \g.
```

```
Your MySQL connection id is 10
```

```
Server version: 8.0.36 MySQL Community Server - GPL
```

```
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```

```
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owners.
```

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> show databases;
```

| Database           |
|--------------------|
| cg_mysql           |
| information_schema |
| performance_schema |

3 rows in set (0.00 sec)

```
mysql> use cg_mysql;
```

```
Database changed
```

```
mysql> show tables;
```

| Tables_in_cg_mysql |
|--------------------|
| students           |

1 row in set (0.00 sec)

```
mysql> select * from students;
```

| student_id | student_name | student_email    |
|------------|--------------|------------------|
| 101        | Praveen      | praveen@cg.com   |
| 102        | jayakumar    | jayakumar@cg.com |
| 103        | sanjay       | sanjay@cg.com    |

```
3 rows in set (0.00 sec)
```

```
mysql> select * from students;
```

| student_id | student_name | student_email    |
|------------|--------------|------------------|
| 101        | Praveen      | praveen@cg.com   |
| 102        | jayakumar    | jayakumar@cg.com |
| 103        | sanjay       | sanjay@cg.com    |
| 104        | ushasri      | usha@cg.com      |

```
4 rows in set (0.00 sec)
```

```
mysql> alter table students drop column student_email;
```

```
ERROR 1142 (42000): ALTER command denied to user 'vuss'@'localhost' for table 'students'
```

```
mysql> drop table students;
```

```
ERROR 1142 (42000): DROP command denied to user 'vuss'@'localhost' for table 'students'
```

```
mysql>
```